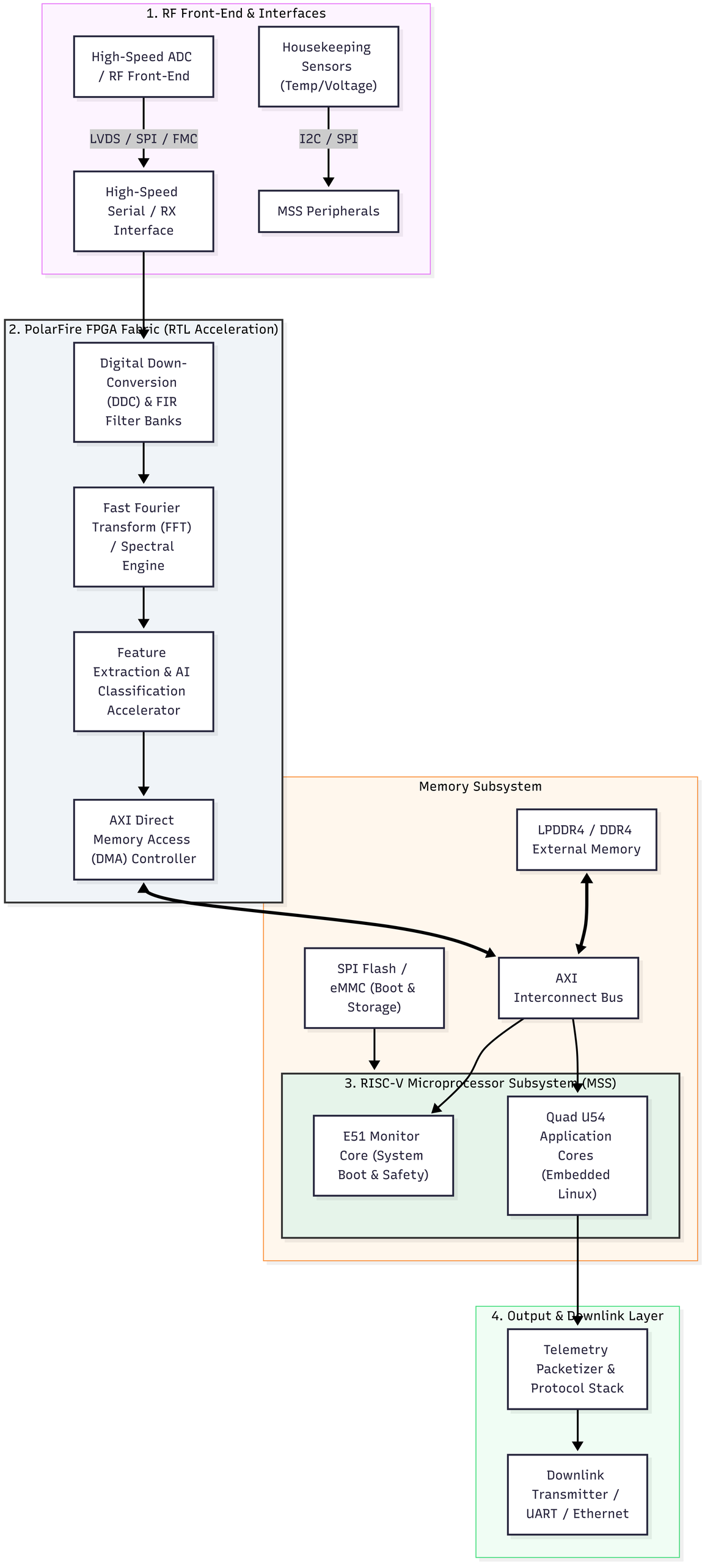

1. RF Front-End & Interfaces
High-Speed ADC / RF Front-End
Housekeeping Sensors (Temp/Voltage)
LVDS / SPI / FMC
I2C / SPI
High-Speed Serial / RX Interface
MSS Peripherals
2. PolarFire FPGA Fabric (RTL Acceleration)
Digital Down-Conversion (DDC) & FIR Filter Banks
Fast Fourier Transform (FFT) / Spectral Engine
Feature Extraction & AI Classification Accelerator
AXI Direct Memory Access (DMA) Controller
Memory Subsystem
LPDDR4 / DDR4 External Memory
SPI Flash / eMMC (Boot & Storage)
AXI Interconnect Bus
3. RISC-V Microprocessor Subsystem (MSS)
E51 Monitor Core (System Boot & Safety)
Quad U54 Application Cores (Embedded Linux)
4. Output & Downlink Layer
Telemetry Packetizer & Protocol Stack
Downlink Transmitter / UART / Ethernet